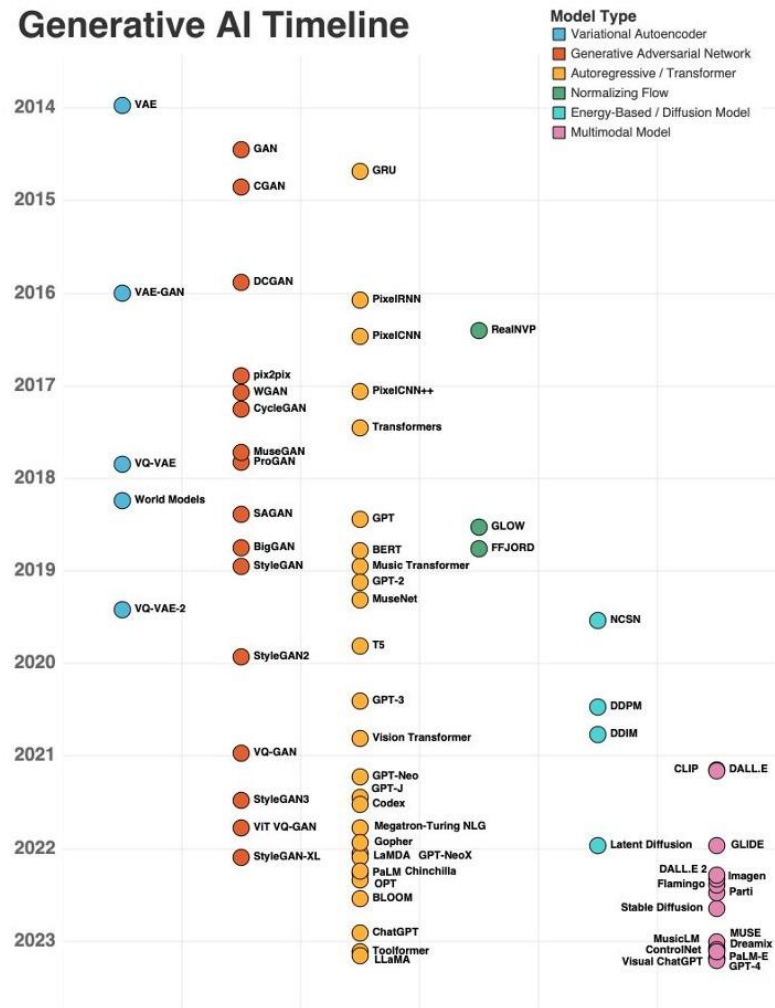


Генеративные модели

Марк Блуменау

О чем речь?

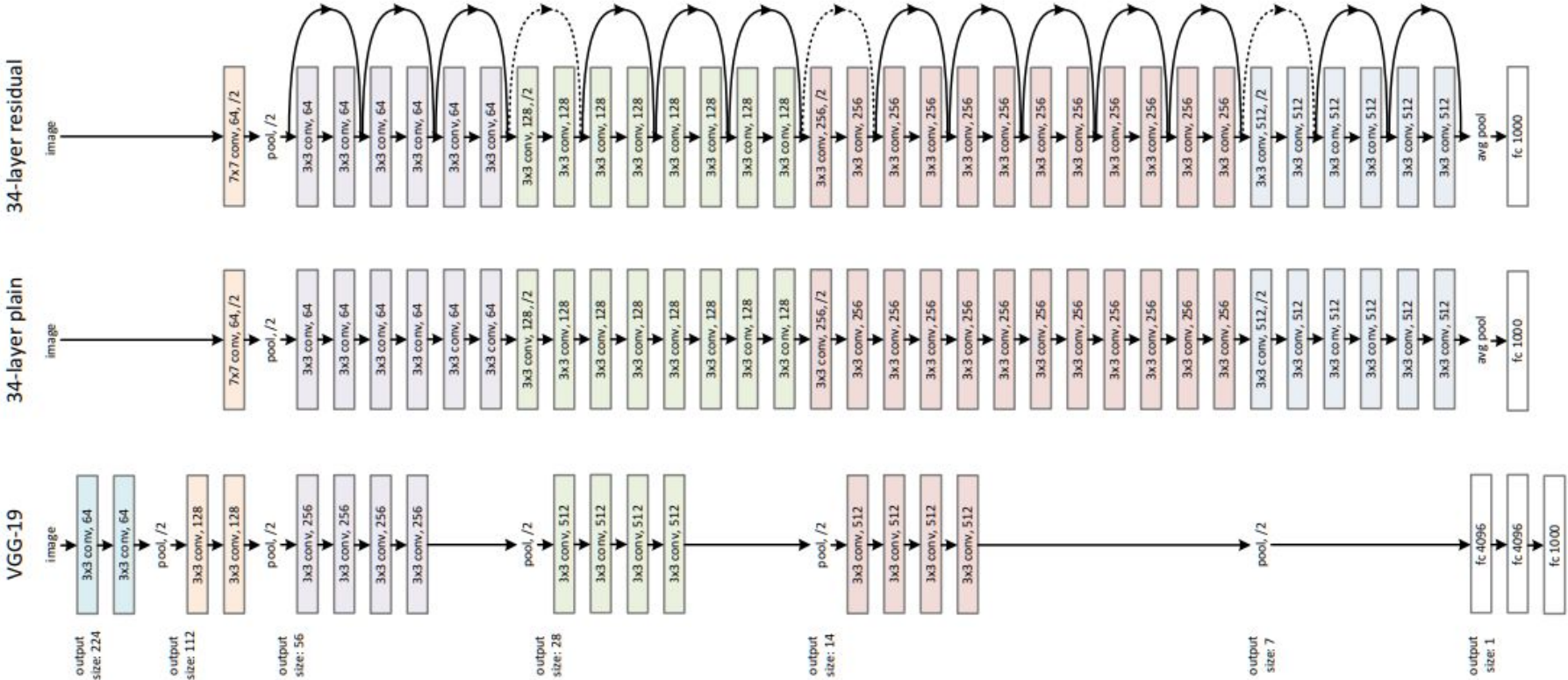
Generative AI Timeline



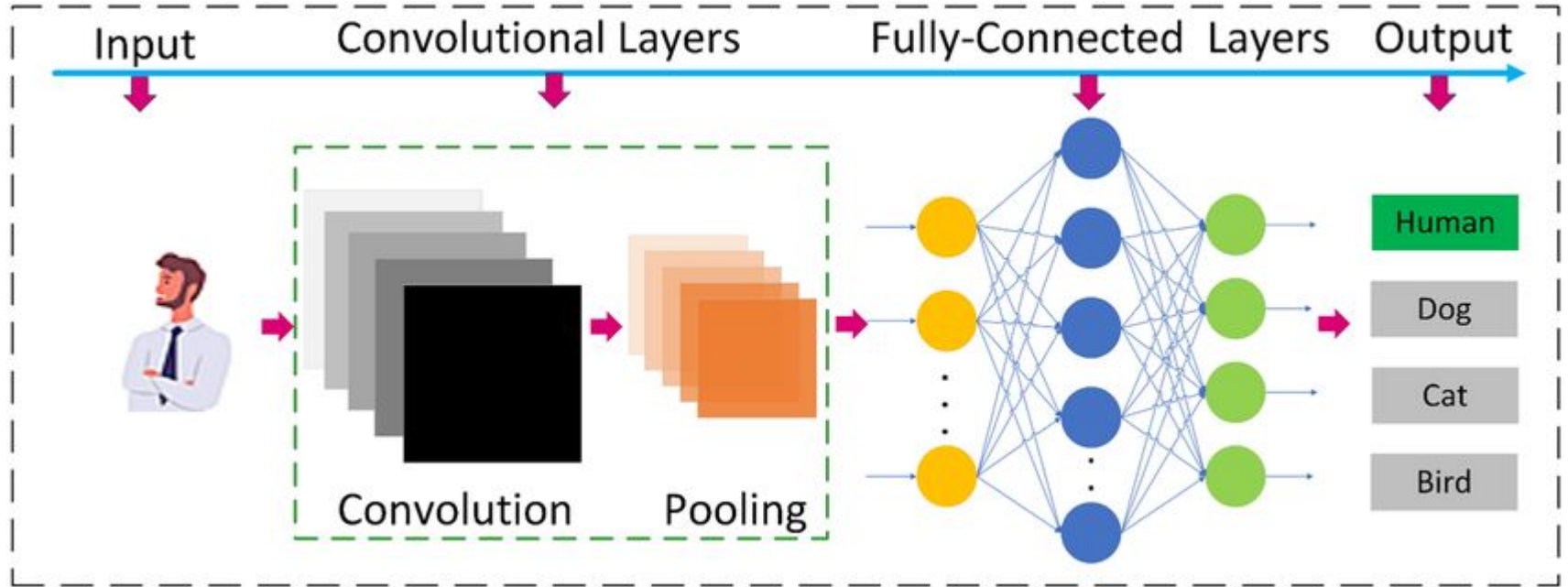


Credit: Elena
Kantonistova

Что мы уже знаем



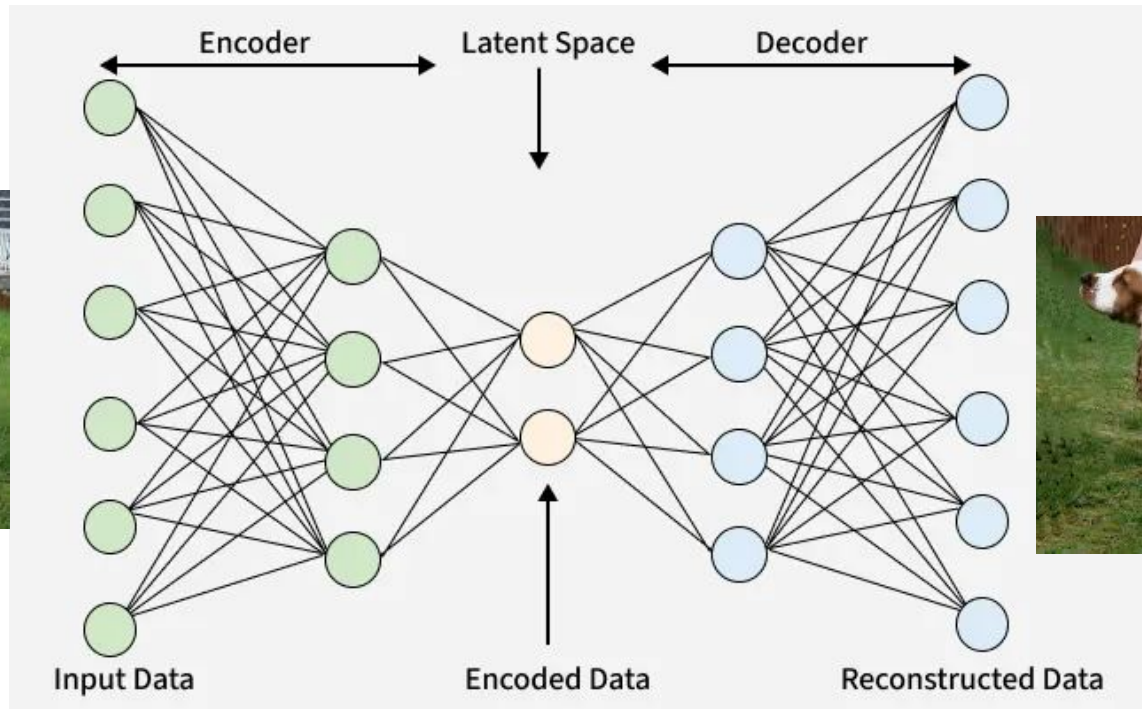
Supervised embeddings



А если разметки нет?

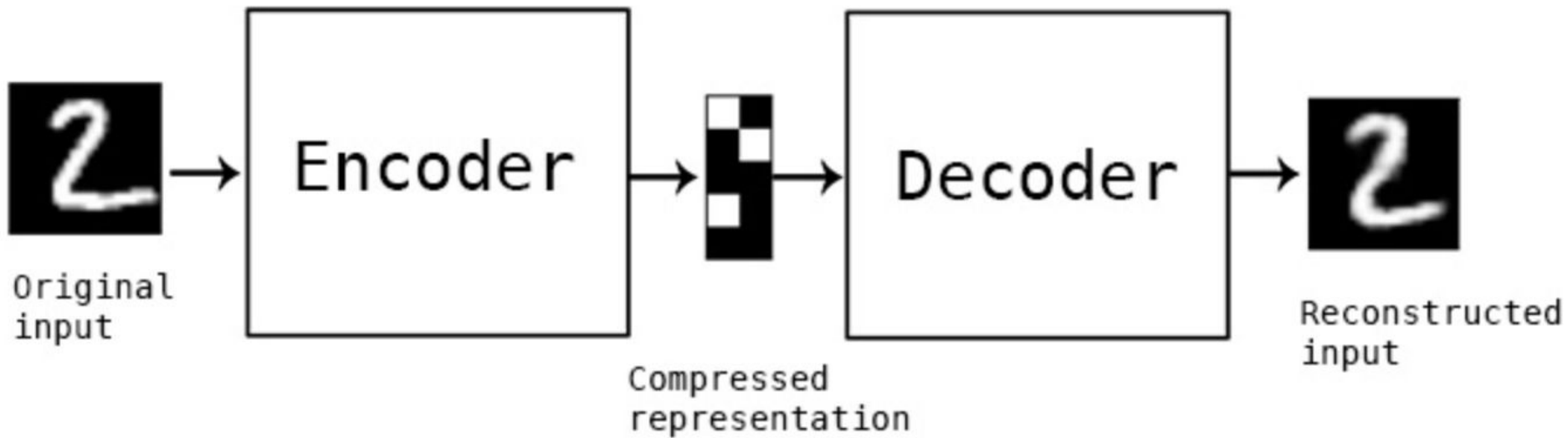
То смотрим старую лекцию?

Нет!

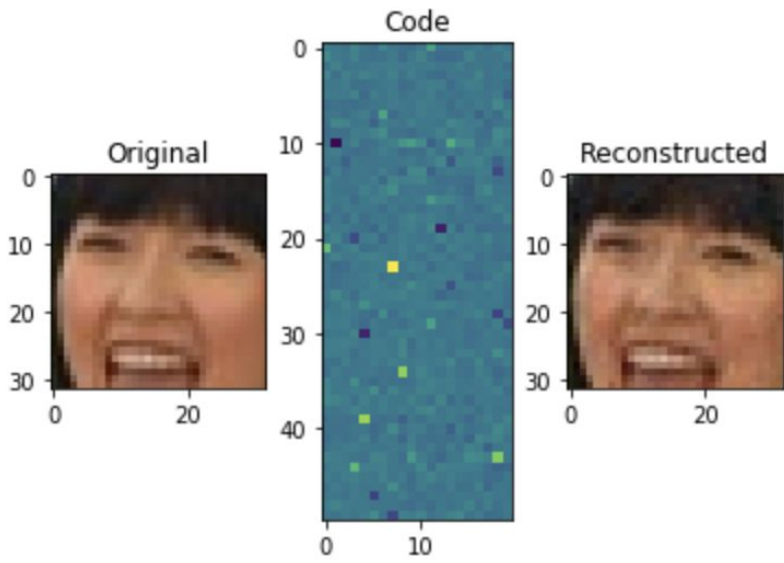
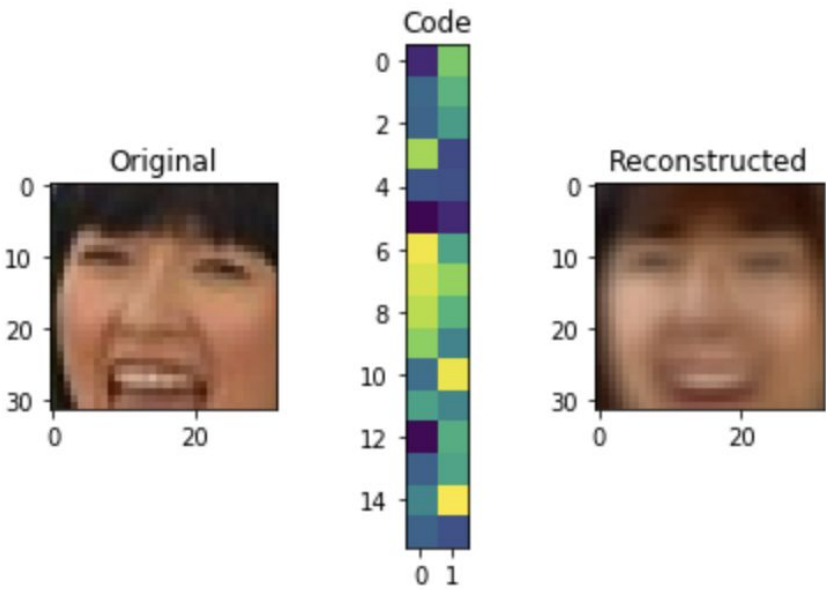


Как учить?

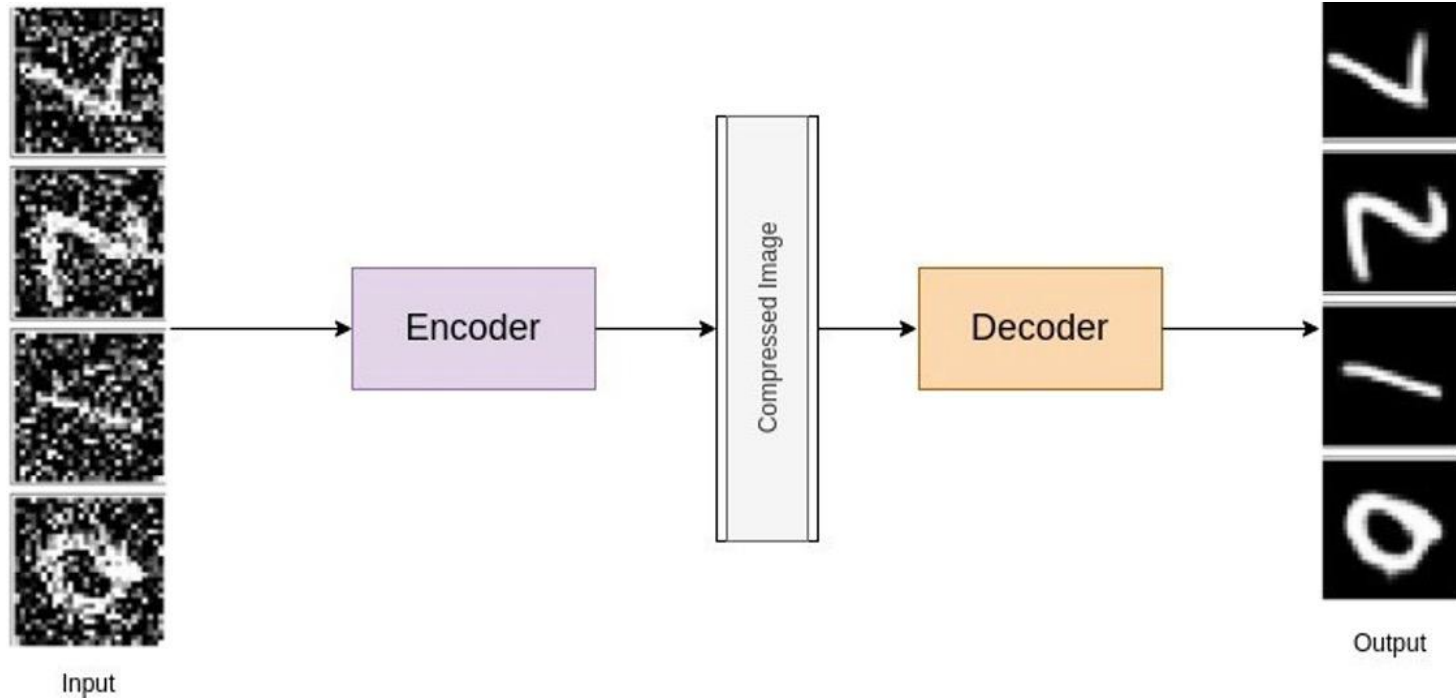
$$L\left(x_i, g\left(f\left(x_i\right)\right)\right) \rightarrow \min$$



Больше – лучше!

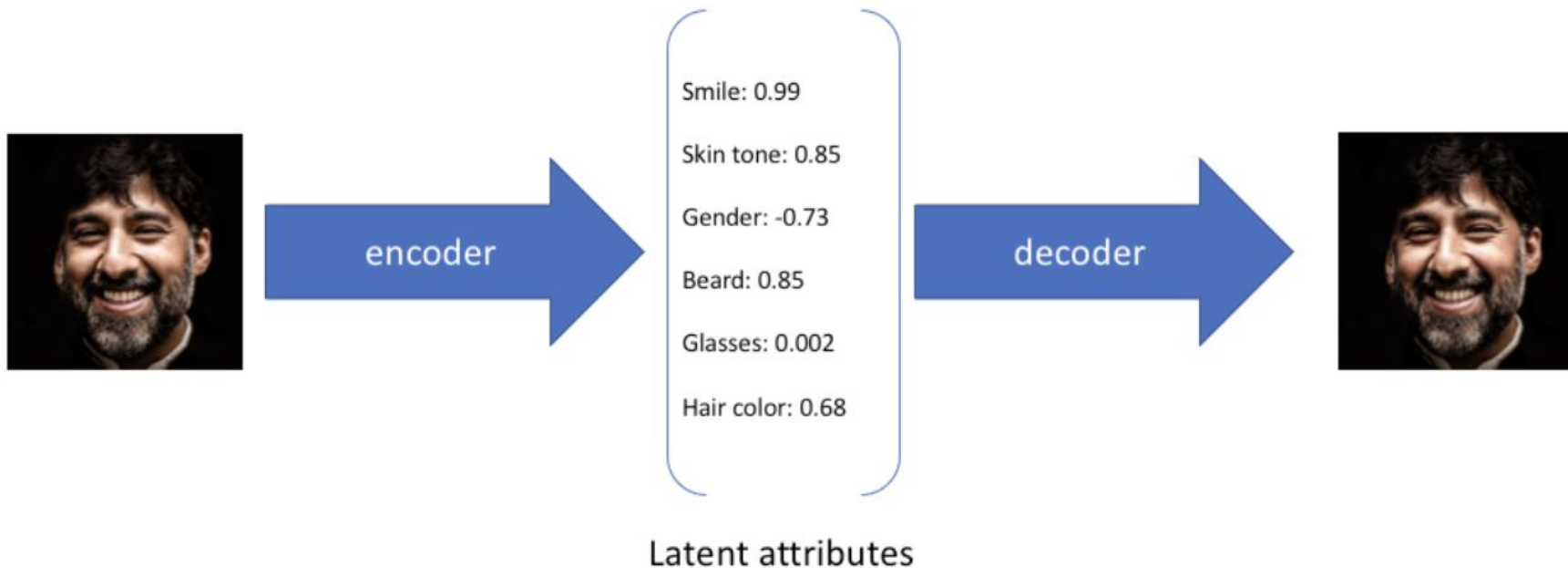


Можно даже такое делать!

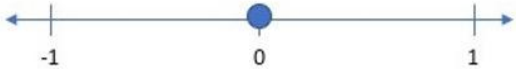
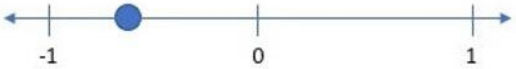


Давайте подумаем зачем оно нам такое надо...

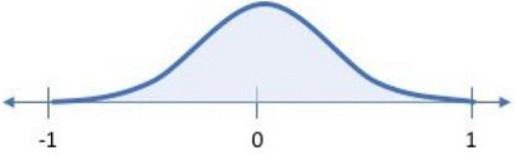
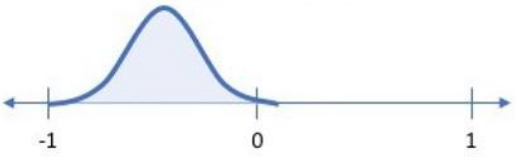
Variational AE = AE but better



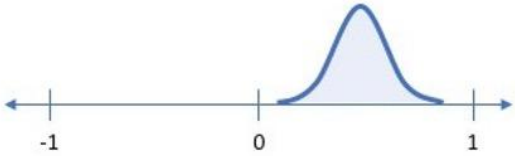
Smile (discrete value)

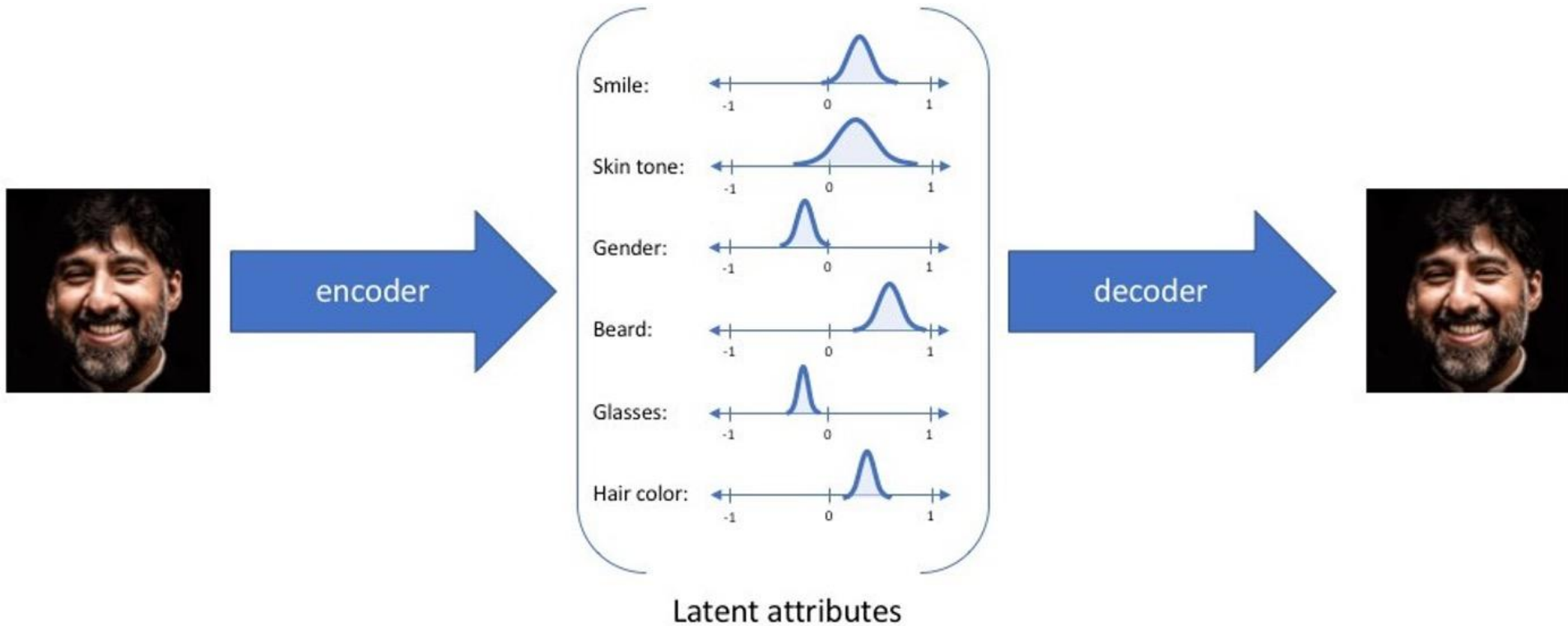


Smile (probability distribution)



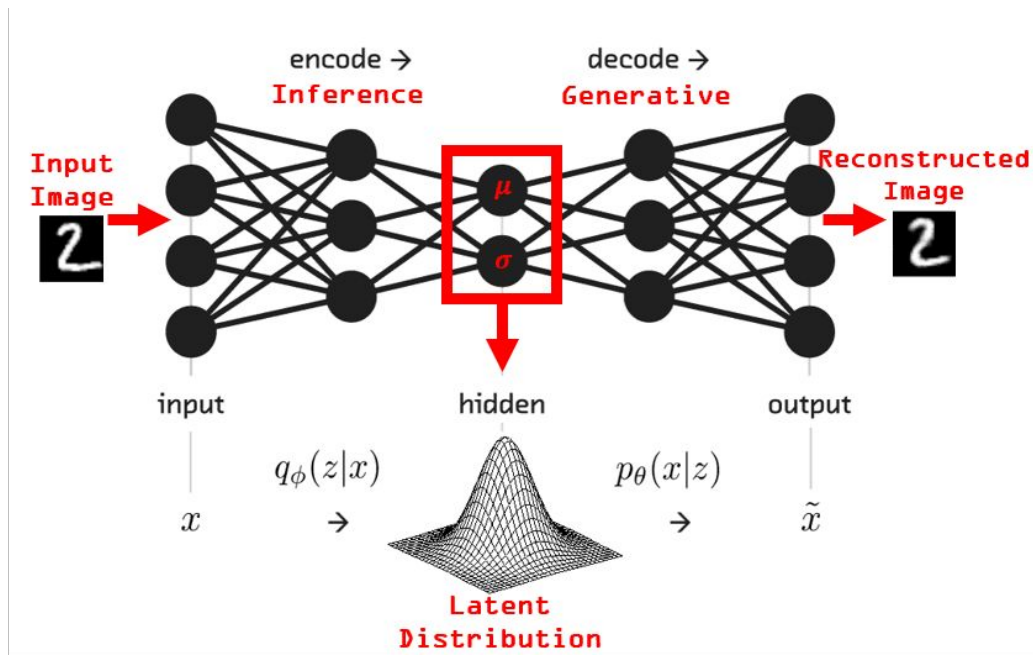
VS.



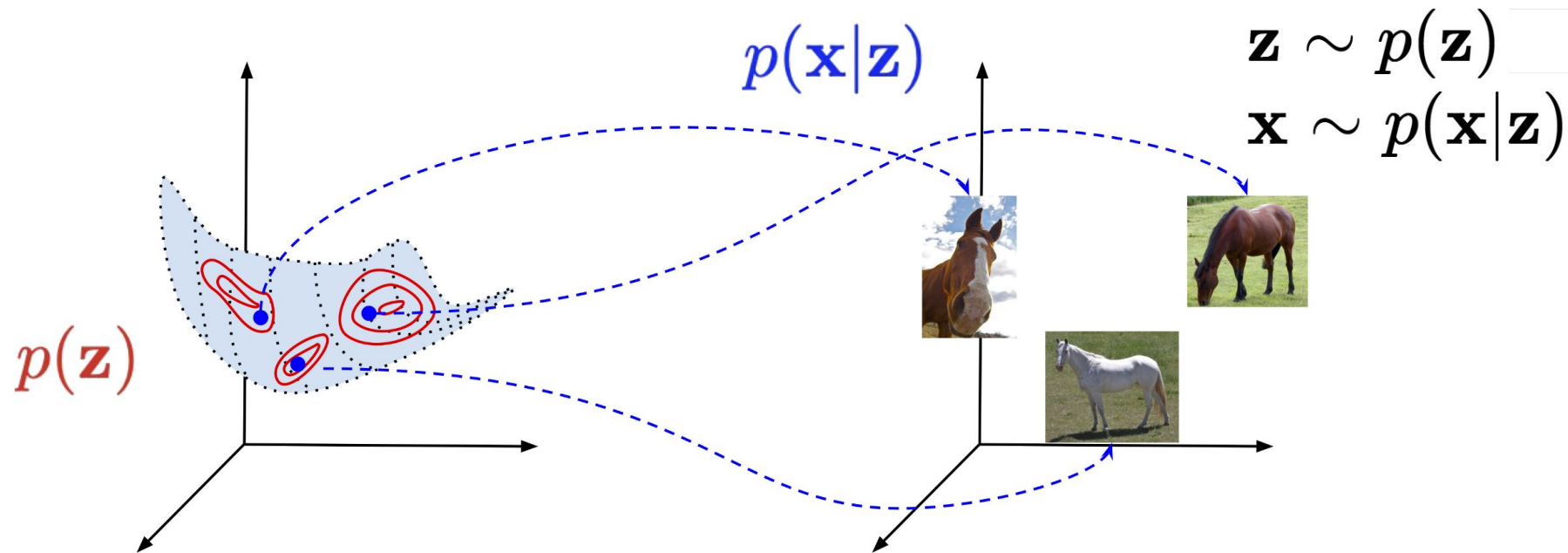


Фактически

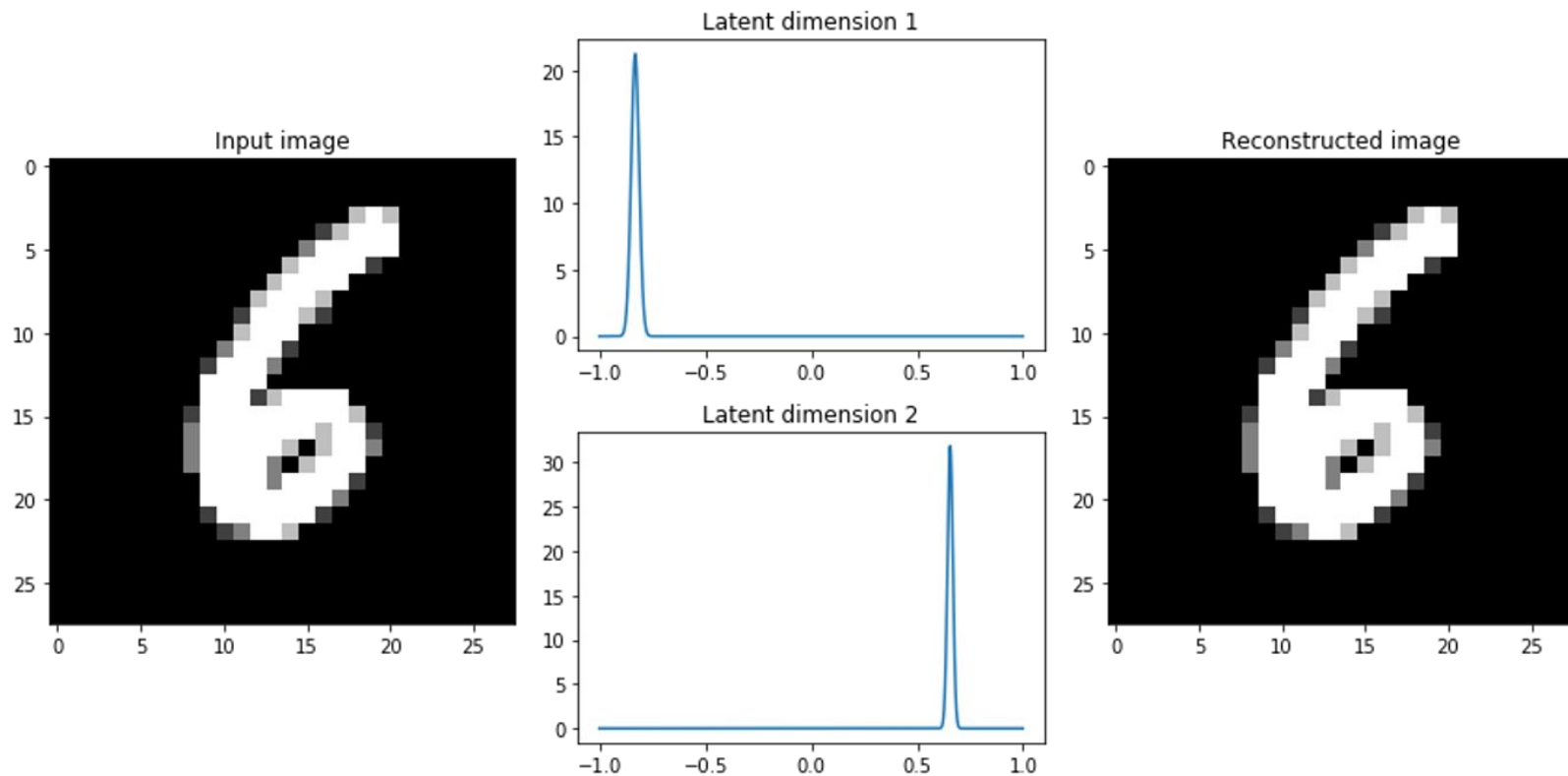
$$\text{encoder}(x) = (\mu(x), \sigma(x))$$



Или вот так



Но как мы не хотим



И как учить?

$$E_{q(z|x)} \log p(x|z) - KL(q(z|x) || p(z)) \rightarrow \max$$

Reconstruction

Regularization

$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}.$$

Реконструкция обычно MSE

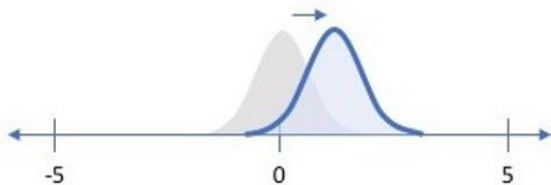
Если decoder выдает среднее гауссианы

$$p(x \mid z) = \mathcal{N}(x \mid \mu_\theta(z), \sigma^2 I)$$

$$\log p(x \mid z) = -\frac{1}{2\sigma^2} \|x - \mu_\theta(z)\|^2 - \frac{d}{2} \log(2\pi\sigma^2)$$

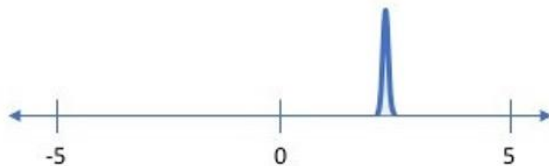
$$-\mathbb{E}_{q(z|x)}[\log p(x|z)] = \frac{1}{2\sigma^2} \mathbb{E}_{q(z|x)} \|x - \mu_\theta(z)\|^2 + \text{const}$$

Penalizing reconstruction loss encourages the distribution to describe the input



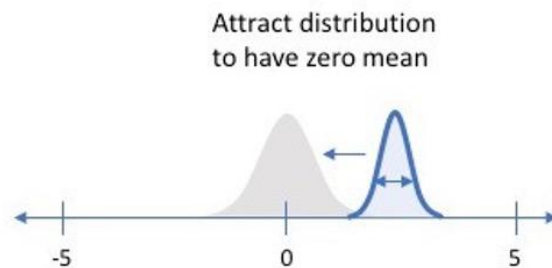
Our distribution deviates from the prior to describe some characteristic of the data

Without regularization, our network can “cheat” by learning narrow distributions



With a small enough variance, this distribution is effectively only representing a single value

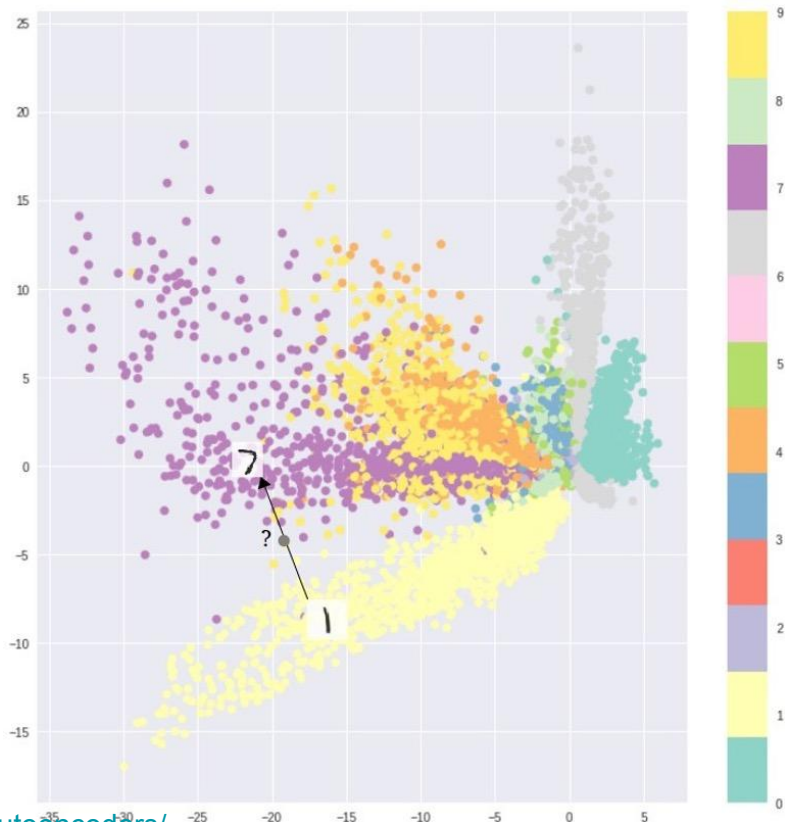
Penalizing KL divergence acts as a regularizing force



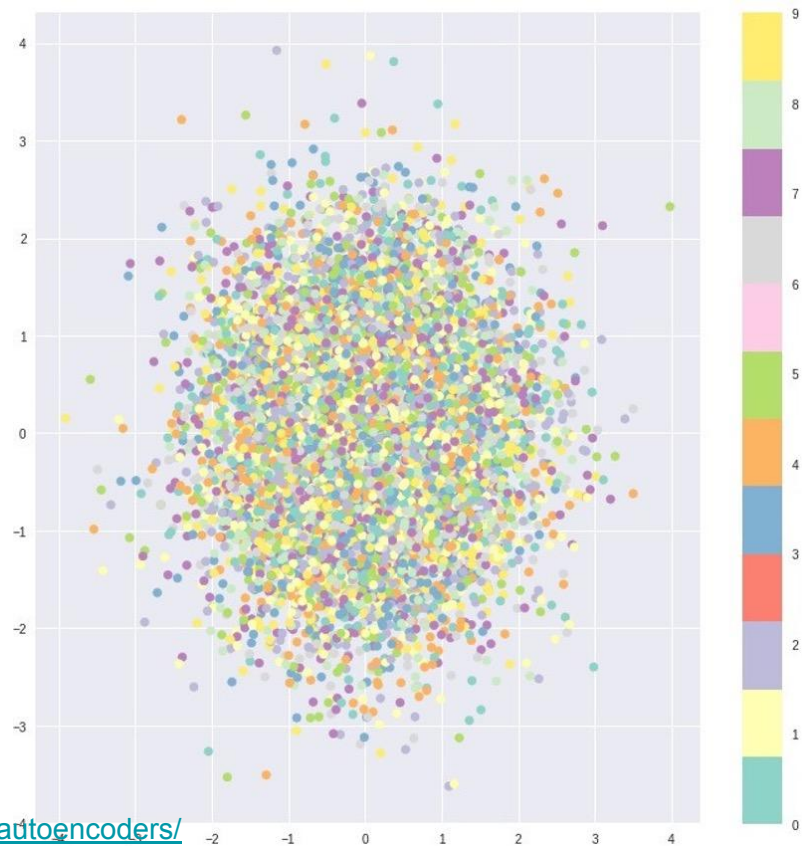
Attract distribution to have zero mean

Ensure sufficient variance to yield a smooth latent space

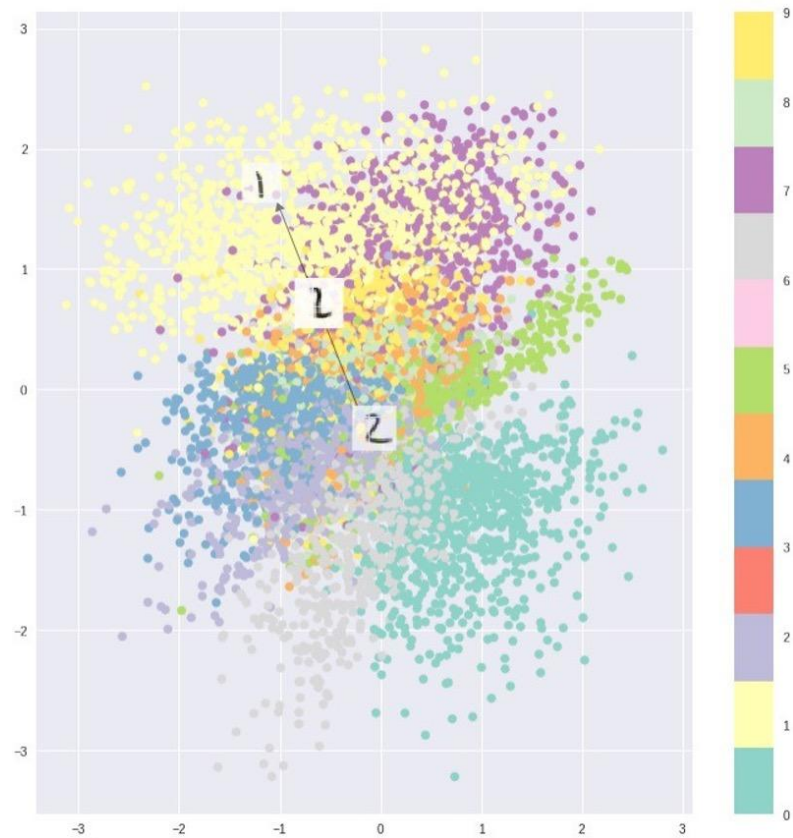
Только reconstruction

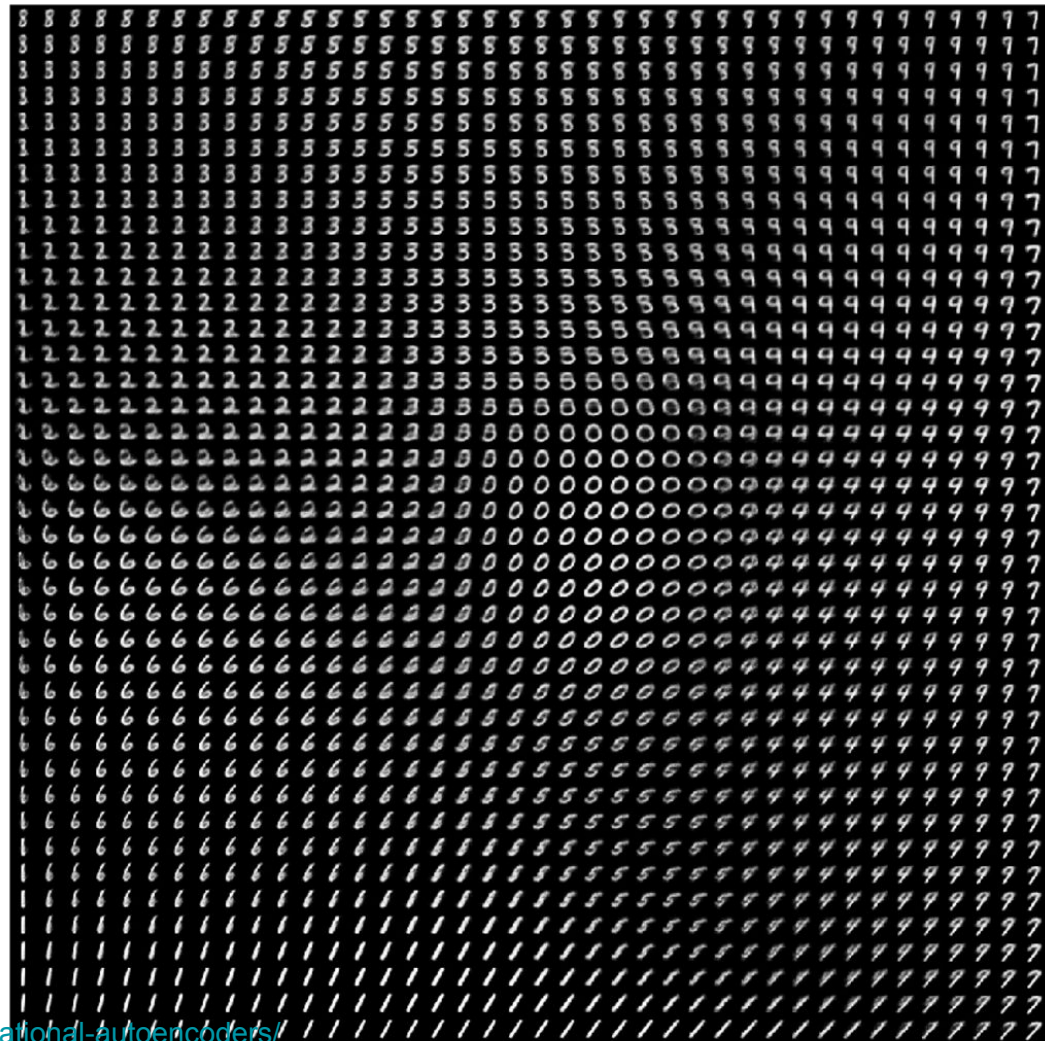


Только KL

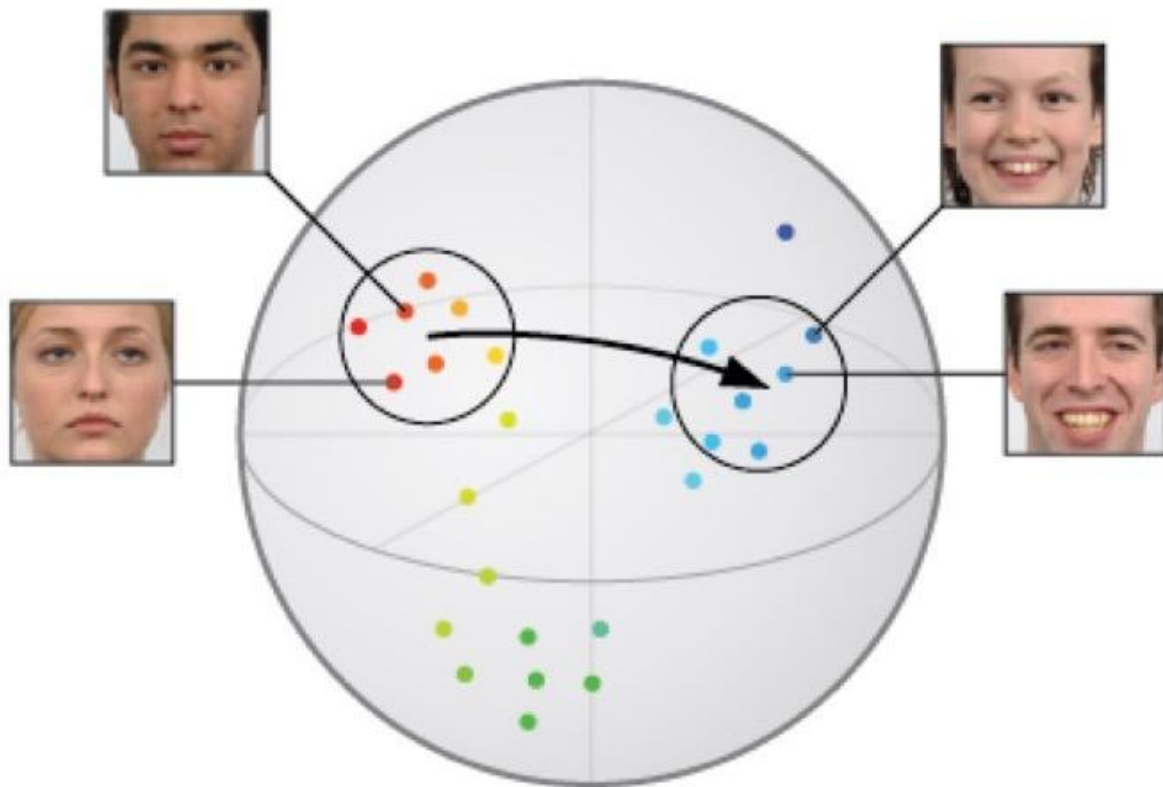


Оба





Улыбочку!



Reconstruction

Gender-biased
smilevector

Gender-balanced
smilevector

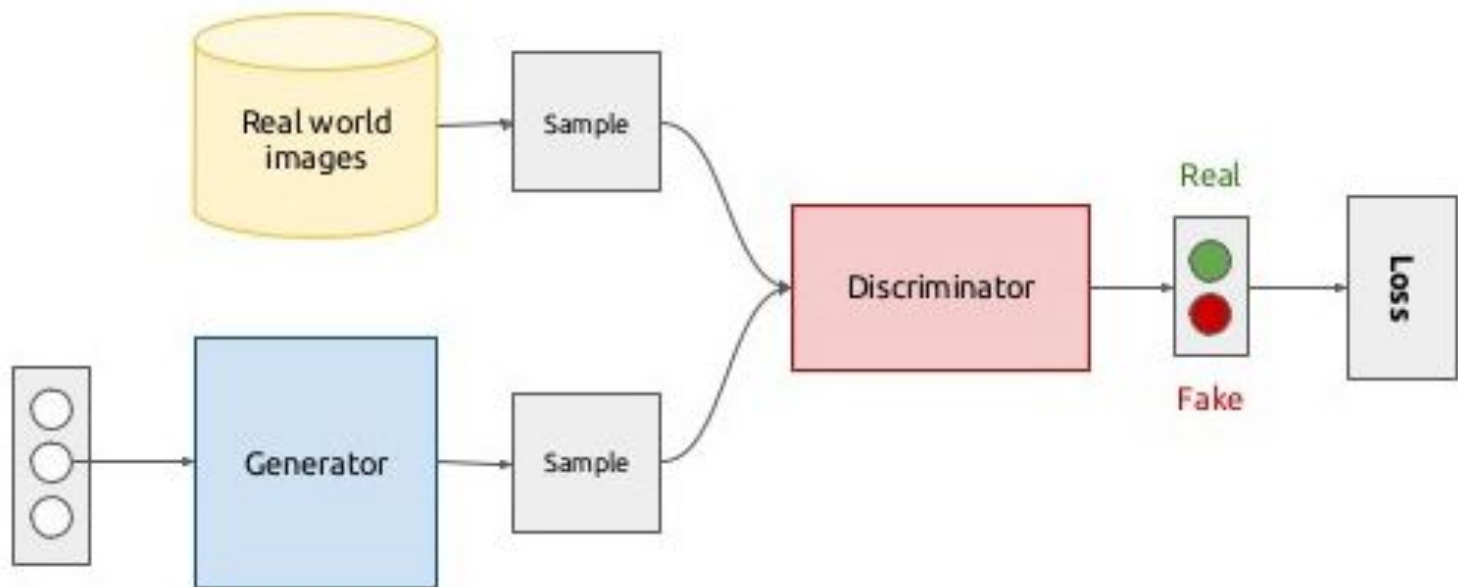


Генерация

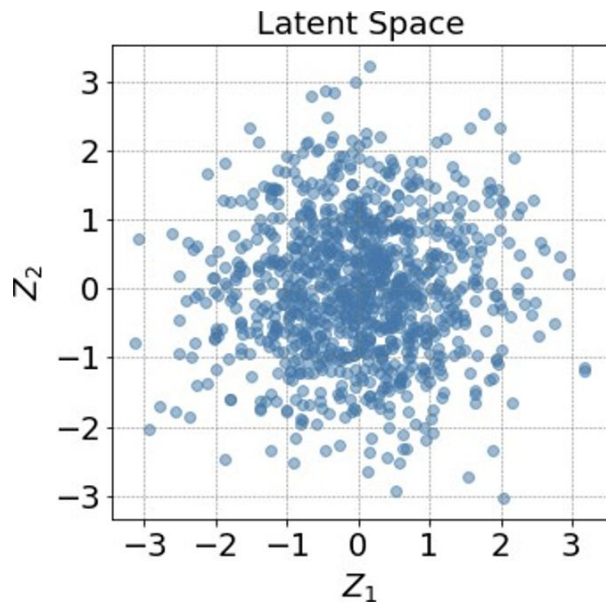


<https://gaussian37.github.io/deep-learning-chollet-8-4/>

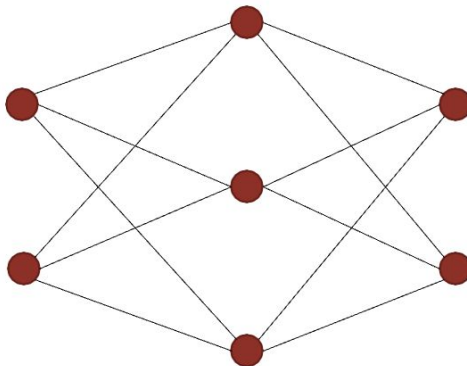
Новый тип модели! GAN



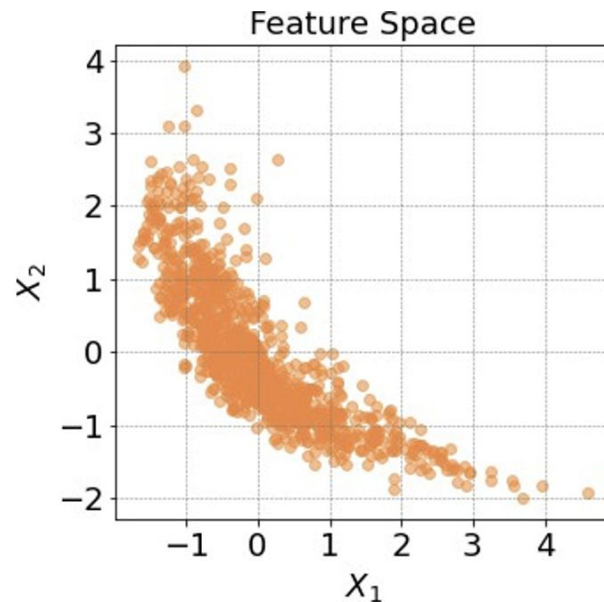
Generator



Z

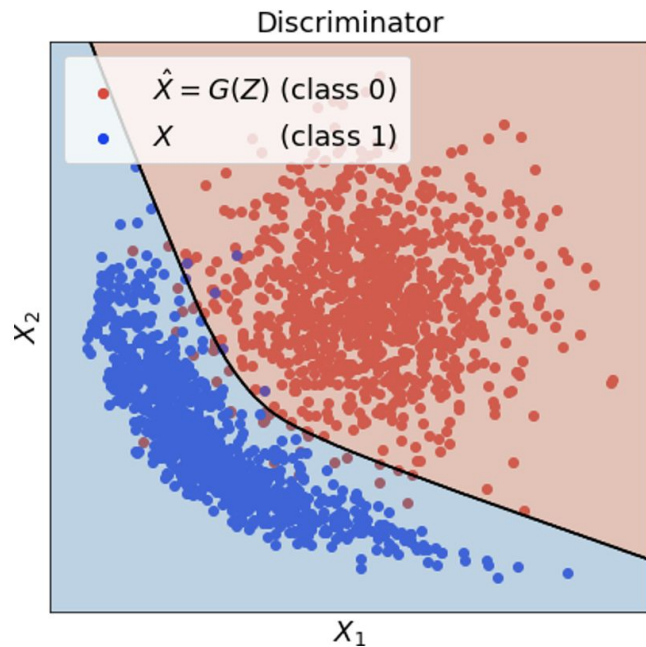


$G(Z)$



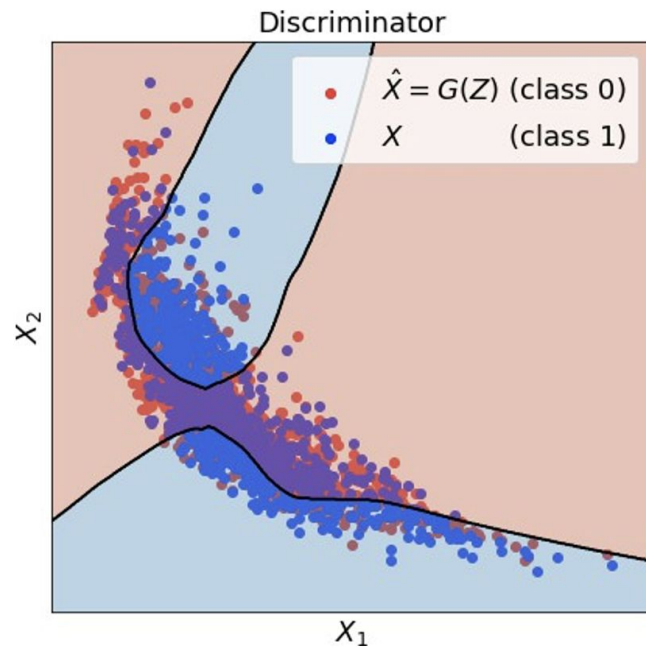
$X = G(Z)$

Discriminator



ROC AUC = 0.995

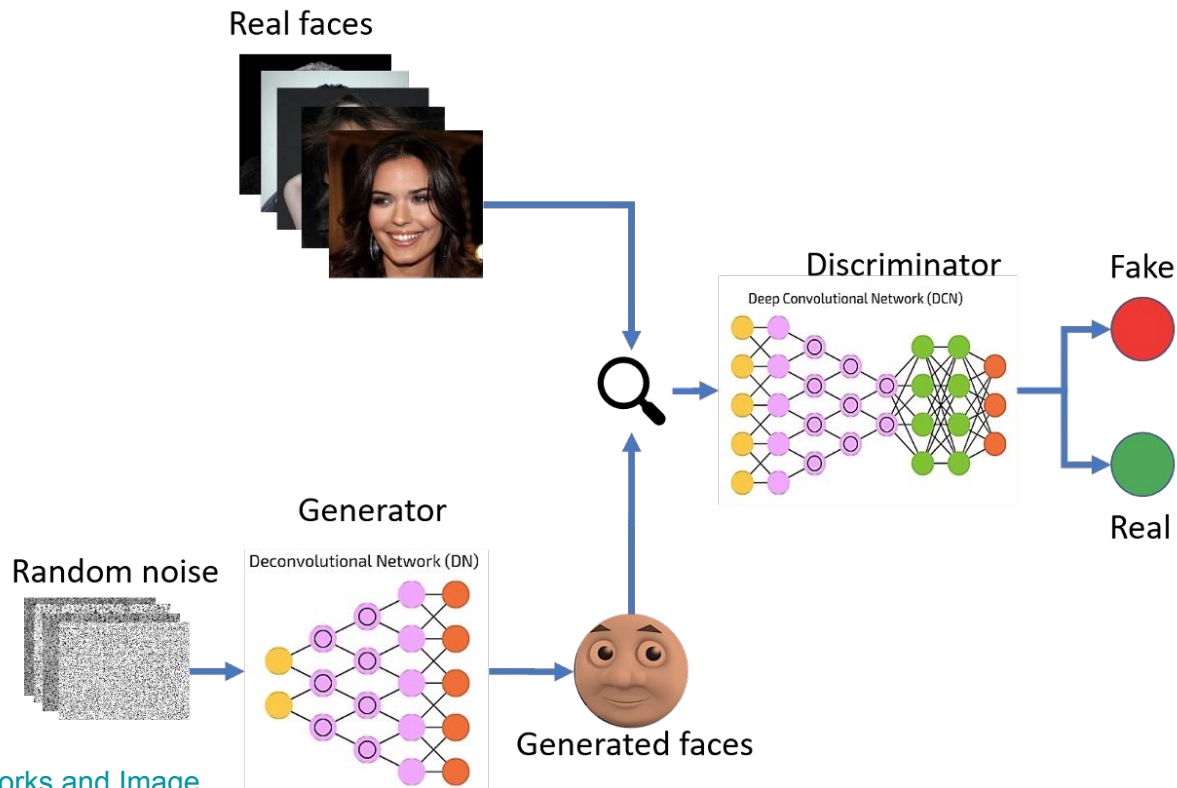
Log Loss = 0.135



ROC AUC = 0.554

Log Loss = 0.689

Суммарно



Как учить?

$$L(\mathbf{D}, \mathbf{G}) = -\frac{1}{n} \sum_{x_i \in X} \log \mathbf{D}(x_i) - \frac{1}{n} \sum_{z_i \in Z} \log(1 - \mathbf{D}(\mathbf{G}(z_i)))$$

$$\max_{\mathbf{G}} \min_{\mathbf{D}} L(\mathbf{D}, \mathbf{G})$$

Результаты



2014



2015



2016

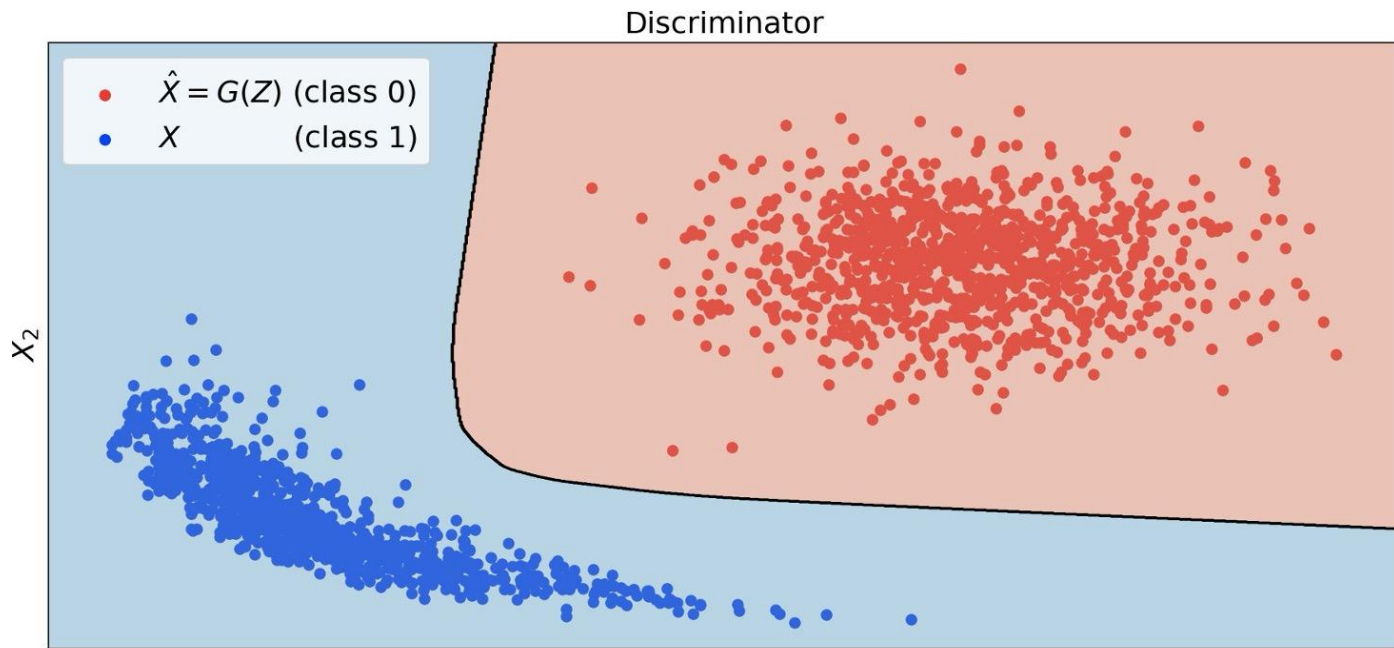


2017



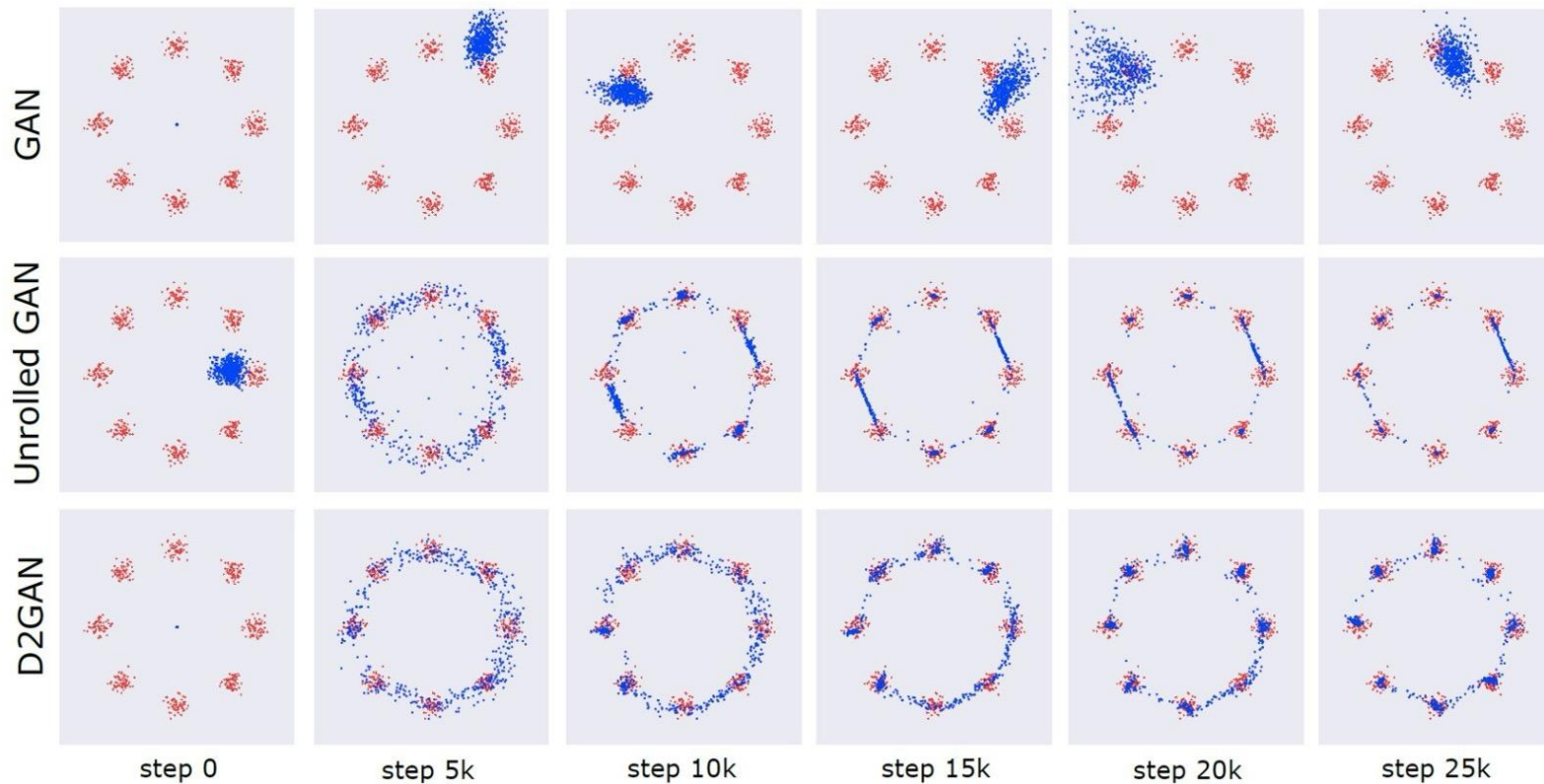
2018

Проблемы

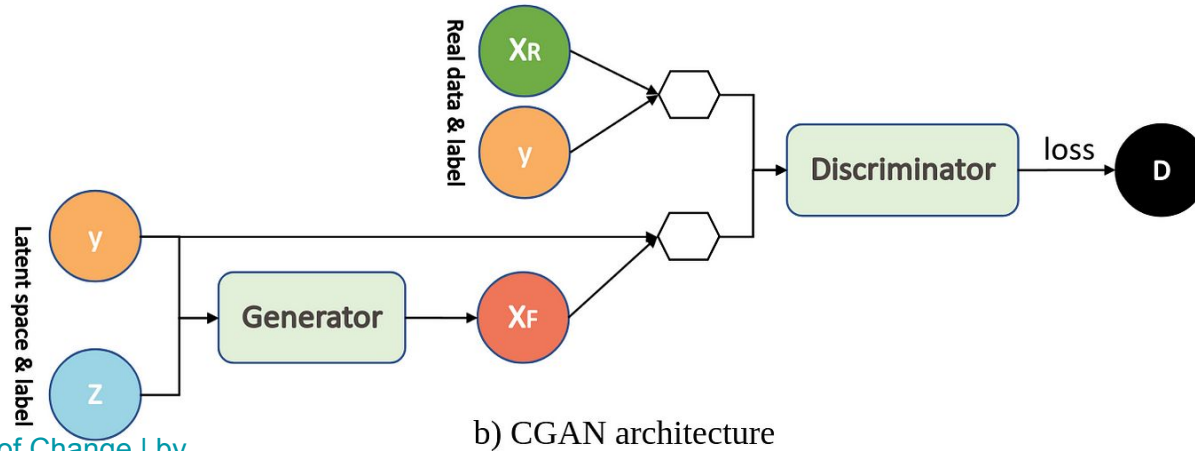
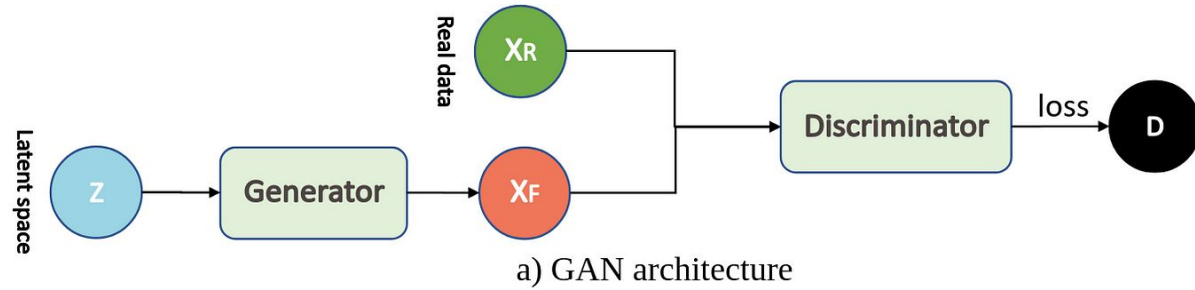


$$L(\mathbf{D}, \mathbf{G}) = -\frac{1}{n} \sum_{x_i \in X} \log \mathbf{D}(x_i) - \frac{1}{n} \sum_{z_i \in Z} \log(1 - \mathbf{D}(\mathbf{G}(z_i)))$$

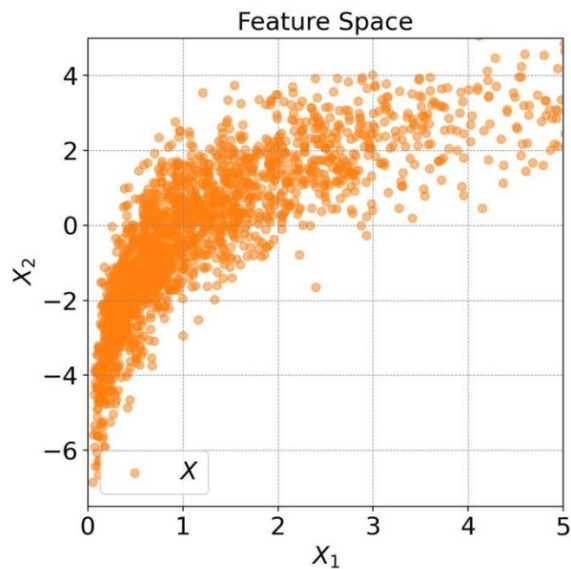
Схлопывание мод



Conditional GAN



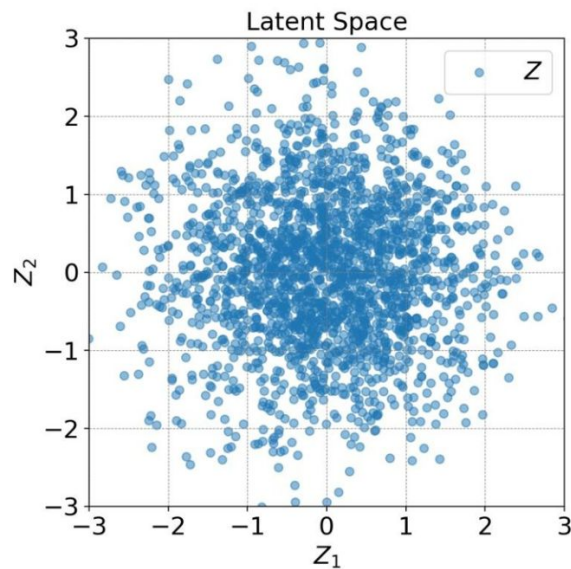
Новый тип модели – Normalizing Flows!



$$x_i \sim p_x(x)$$

$$p_x(x) - ?$$

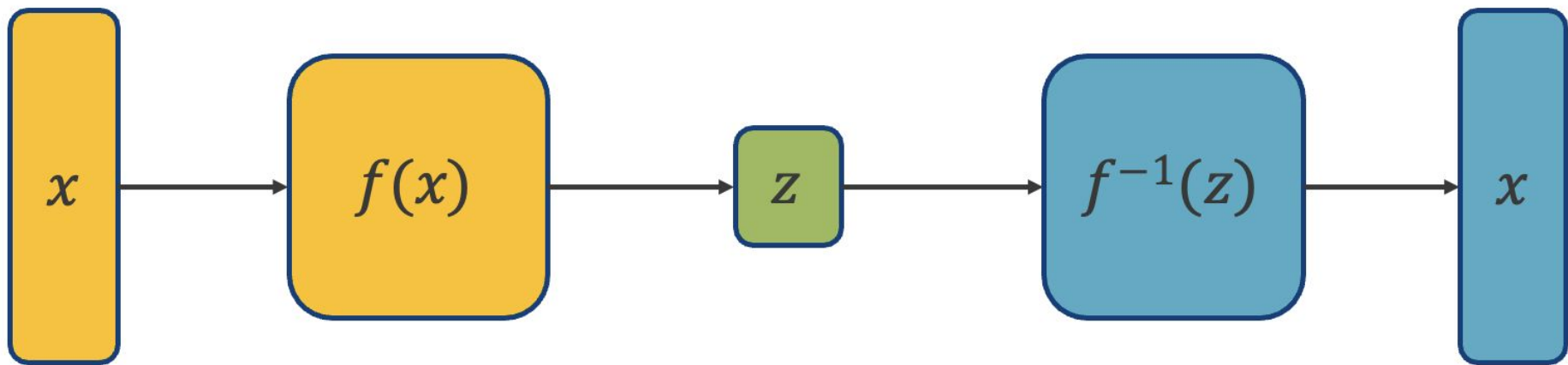
$$z = f(x) - ?$$



$$z_i \sim p_z(z)$$

$$p_z(z) - \text{known}$$

Модель



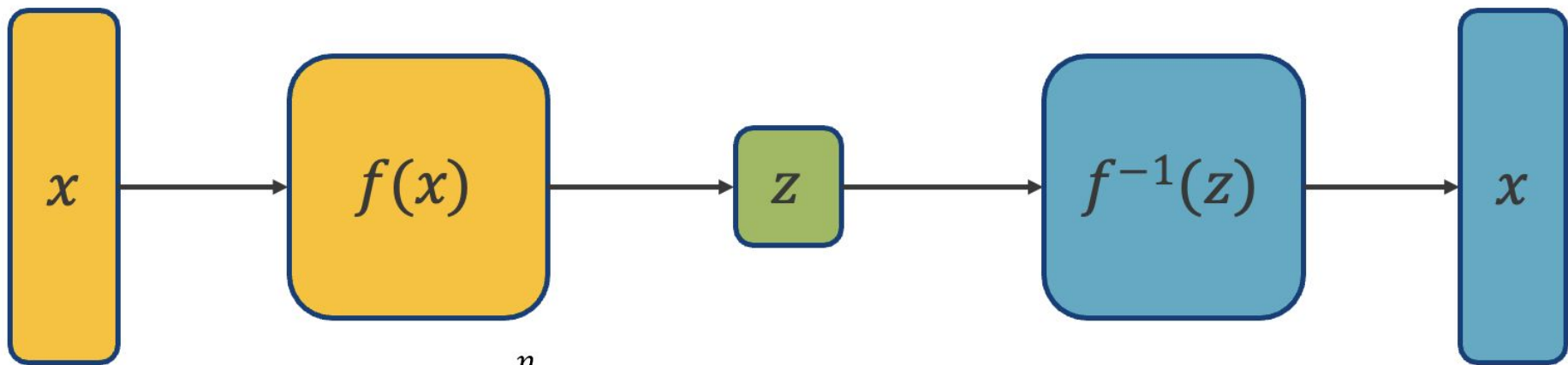
Как учить?

$$L = -\frac{1}{n} \sum_{i=1}^n \log p_x(x_i)$$

$$p_x(x_i) = p_z(f(x_i)) \left| \det \frac{\partial f(x_i)}{\partial x_i} \right|$$

$$L = -\frac{1}{n} \sum_{i=1}^n \left(\log p_z(f(x_i)) + \log \left| \det \frac{\partial f(x_i)}{\partial x_i} \right| \right)$$

Модель



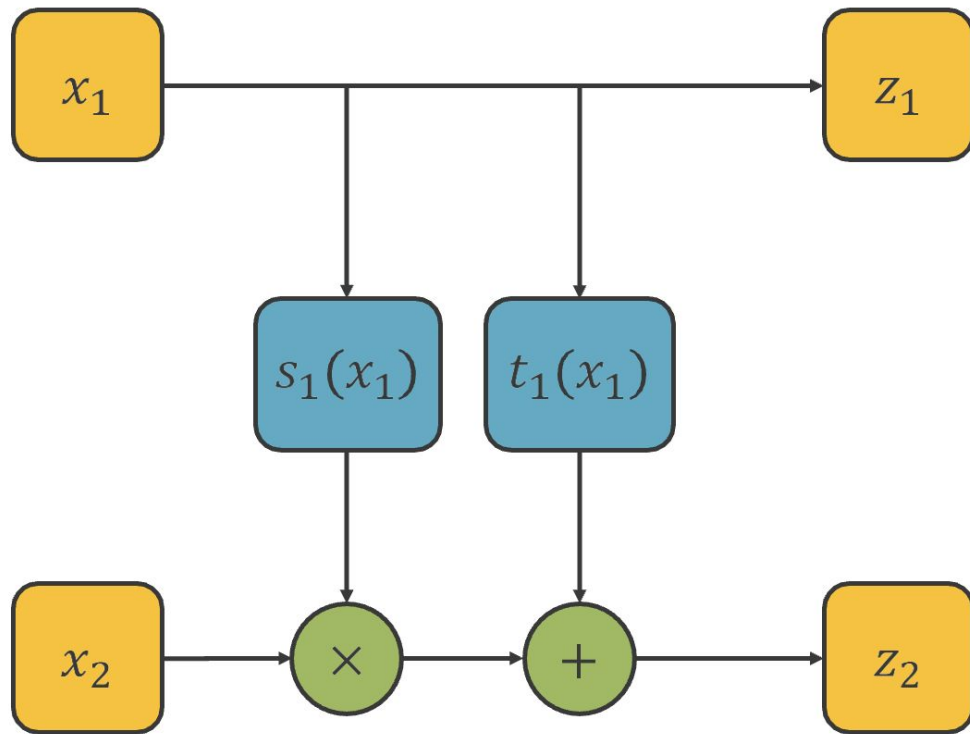
$$L = -\frac{1}{n} \sum_{i=1}^n \left(\log p_z(f(x_i)) + \log \left| \det \frac{\partial f(x_i)}{\partial x_i} \right| \right)$$

Но!

Функция должна быть:

- 1) Дифференцируемая
- 2) Обратимая

Real NVP



Real NVP

s, t – нейросети

$$z = \mathbf{f}(x) = \begin{cases} z_{1:d} = x_{1:d} \\ z_{d+1:D} = x_{d+1:D} \odot \exp(\mathbf{s}(x_{1:d})) + \mathbf{t}(x_{1:d}) \end{cases}$$

$$x = \mathbf{f}^{-1}(z) = \begin{cases} x_{1:d} = z_{1:d} \\ x_{d+1:D} = (z_{d+1:D} - \mathbf{t}(x_{1:d})) \odot \exp(-\mathbf{s}(x_{1:d})) \end{cases}$$

Математика

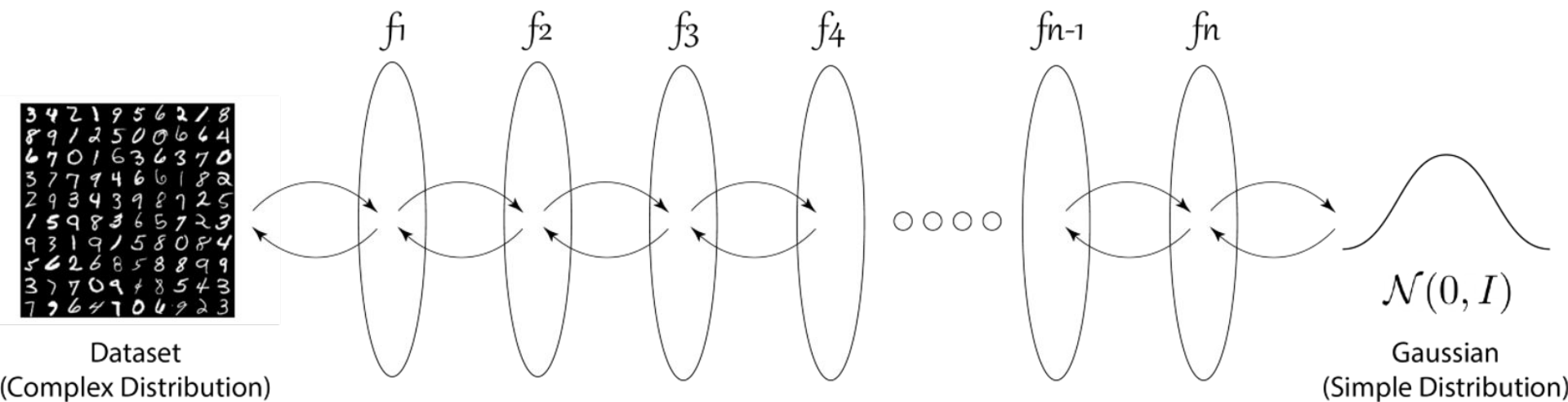
$$\frac{\partial \mathbf{f}(x)}{\partial x} = \begin{pmatrix} \mathbb{I}_d & 0 \\ \frac{\partial z_{1:d}}{\partial x_{1:d}} & \text{diag}(\exp(\mathbf{s}(x_{1:d}))) \end{pmatrix}$$

$$\left| \det \frac{\partial \mathbf{f}(x)}{\partial x} \right| = \exp\left(\sum_{j=d+1}^D \mathbf{s}(x_{1:d})_j \right)$$

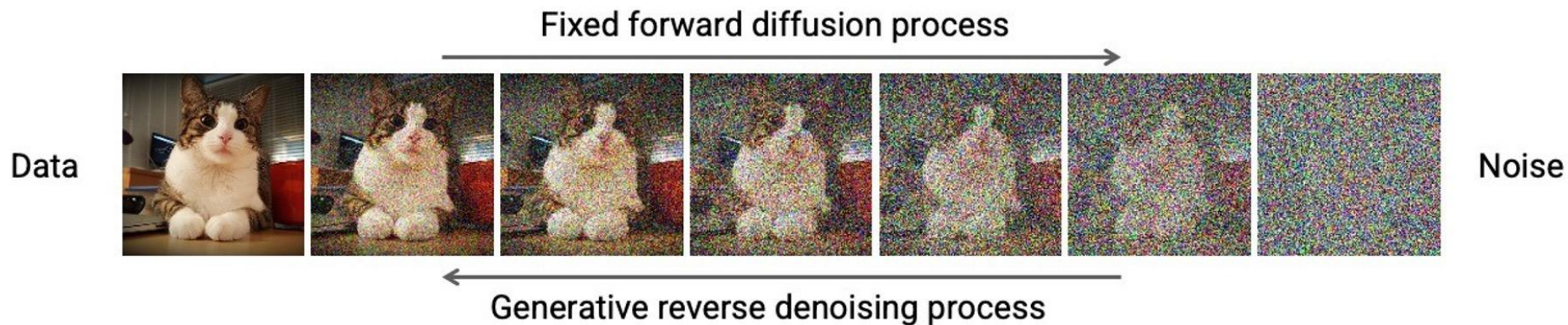
Результаты



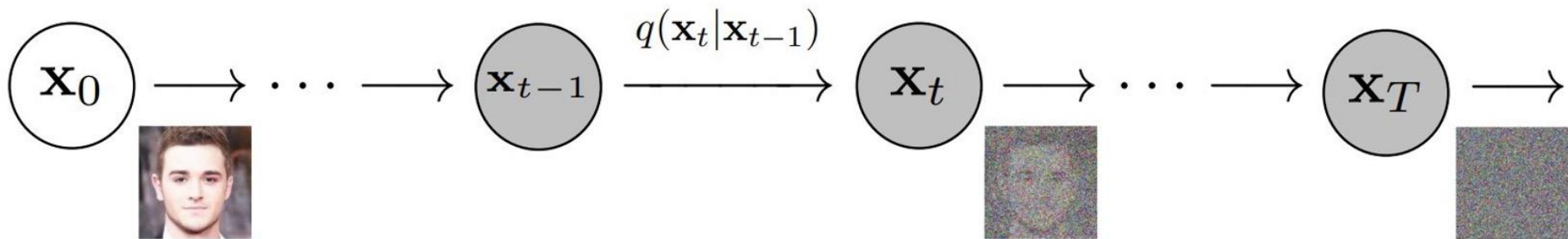
Normalizing flow in a nutshell



Новая модель – Diffusion!

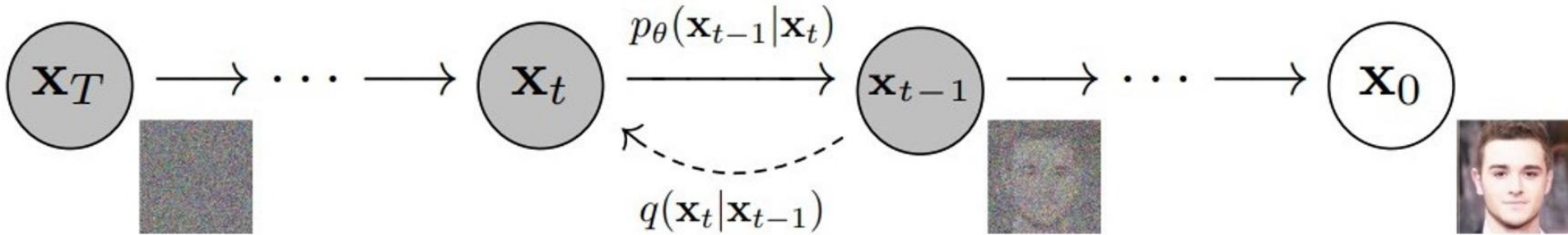


Посолите, поперчите по вкусу

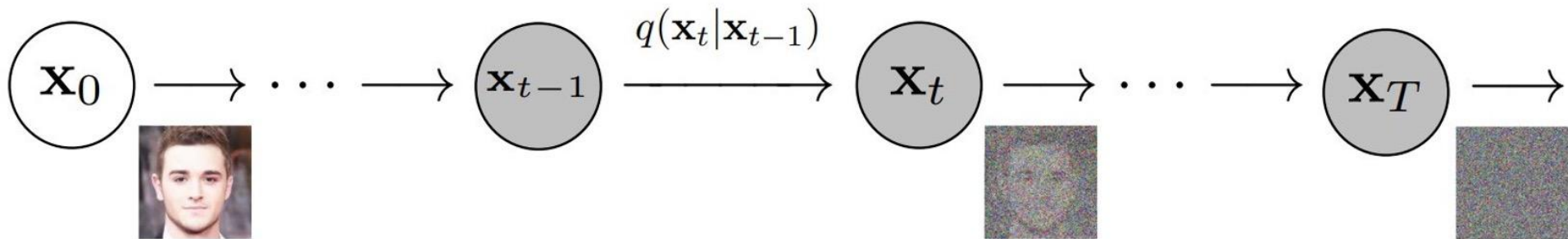


$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \boldsymbol{\Sigma}_t = \beta_t \mathbf{I})$$

Уберите лишнюю соль и перец



Как меняется изображение?



$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \boldsymbol{\Sigma}_t = \beta_t \mathbf{I})$$

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \epsilon_t$$

$$\epsilon_t \sim N(0, \mathbf{I}), \quad \alpha_t = 1 - \beta_t$$

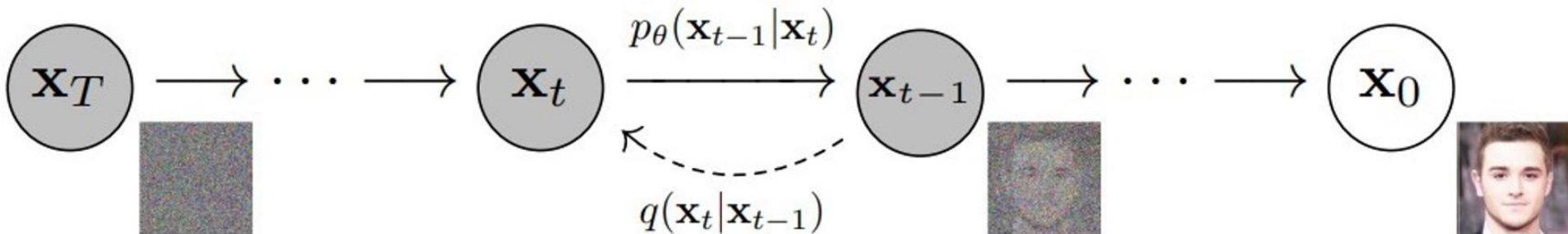
$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

$$\epsilon \sim N(0, \mathbf{I}), \quad \bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right) + \tilde{\beta}_t \mathbf{z}$$

$$\mathbf{z} \sim N(0, \mathbf{I}), \quad \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

Как расколдовывается изображение?



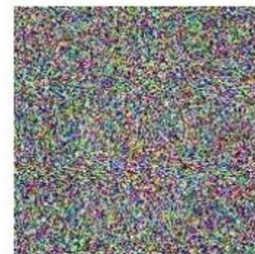
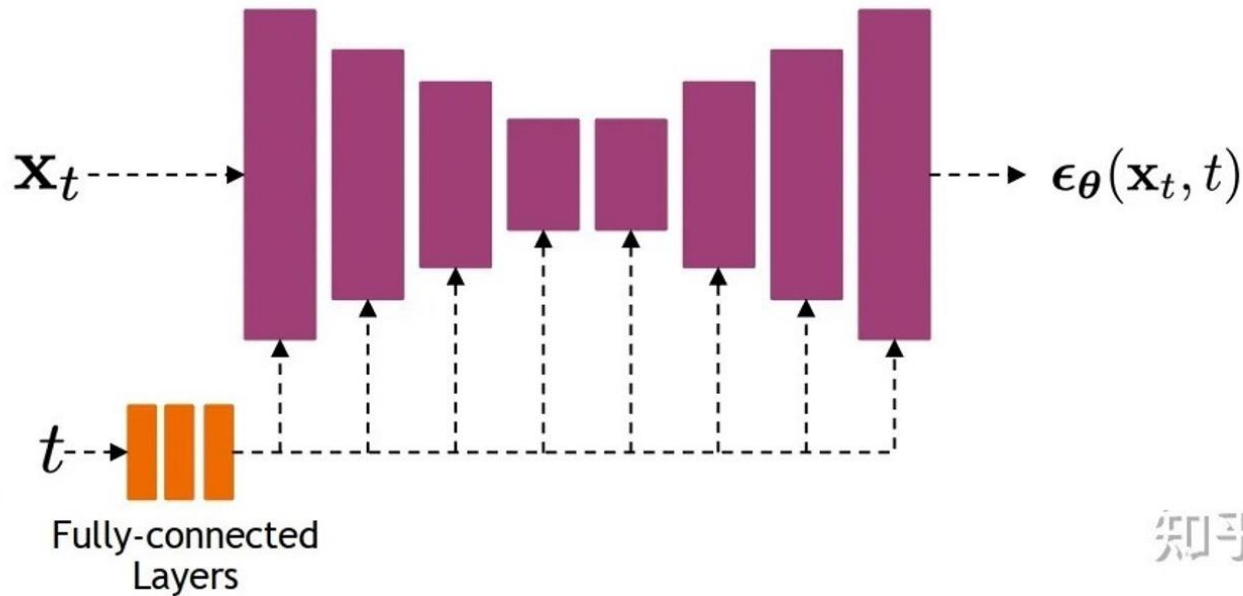
$$\hat{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z \quad z \sim N(0, I), \quad \sigma_t = \text{const}$$

$$L_t = \|x_{t-1} - \hat{x}_{t-1}\|_2^2 \propto \|\epsilon - \epsilon_\theta(x_t, t)\|_2^2 \rightarrow \min_{\theta}$$

А еще хорошо время подавать



Time Representation



知乎 @华年SS

Алгоритм обучения

Algorithm 1 Training

1: **repeat**

2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$

3: $t \sim \text{Uniform}(\{1, \dots, T\})$

4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

5: Take gradient descent step on

$$\nabla_{\theta} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, t) \right\|^2$$

6: **until** converged

Алгоритм генерации

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
```

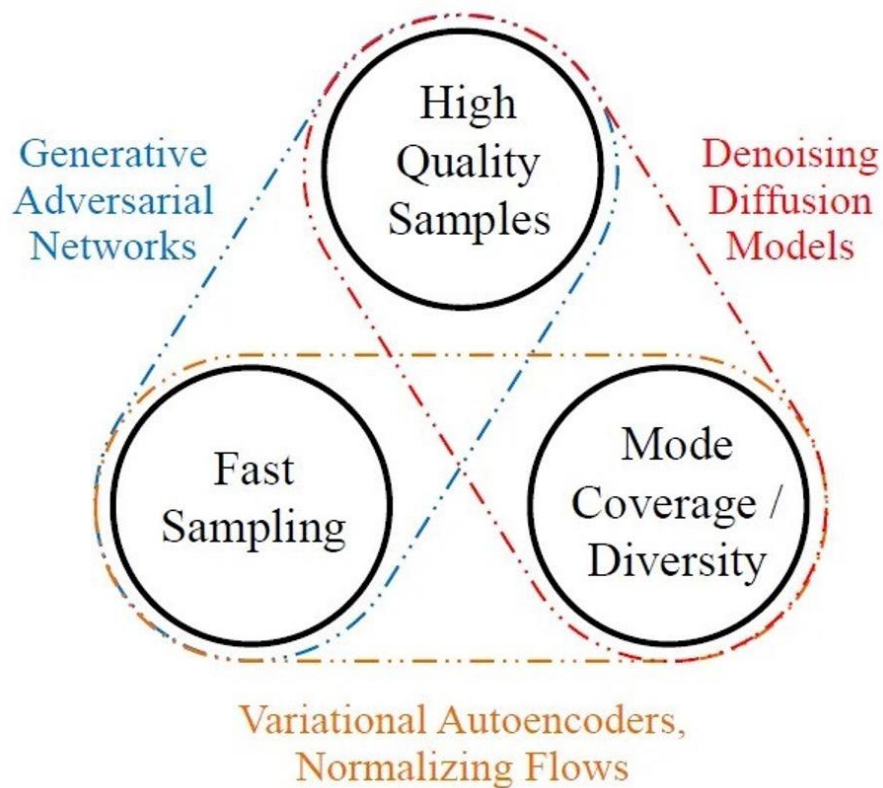

Scheduling

$$T = 1000$$

$$\beta_1 < \beta_2 < \dots < \beta_t < \dots < \beta_T$$

$$\beta_1 = 0.0001, \beta_T = 0.02$$

Трилемма



VAE Math

$$\mathbf{z} \sim p(\mathbf{z})$$

$$\mathbf{x} \sim p(\mathbf{x}|\mathbf{z})$$

$$\begin{aligned} p(\mathbf{x}) &= \int p(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) \, d\mathbf{z} \\ &= \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} [p(\mathbf{x}|\mathbf{z})] \\ &\approx \frac{1}{K} \sum_k p(\mathbf{x}|\mathbf{z}_k) \end{aligned}$$

But $\mathbf{z} \in \mathbb{R}^M$

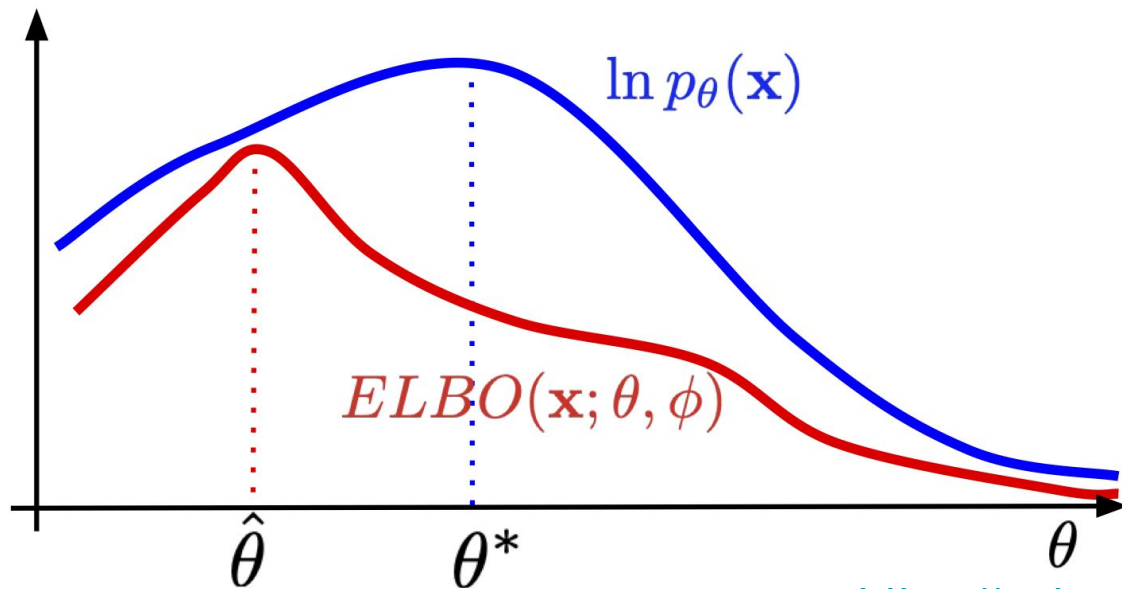
Oh no, we have a problem!

VAE

$$\begin{aligned}\ln p(\mathbf{x}) &= \mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} [\ln p(\mathbf{x})] \\&= \mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\ln \frac{p(\mathbf{z}|\mathbf{x})p(\mathbf{x})}{p(\mathbf{z}|\mathbf{x})} \right] \\&= \mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\ln \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p(\mathbf{z}|\mathbf{x})} \right] \\&= \mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\ln \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p(\mathbf{z}|\mathbf{x})} \frac{q_{\phi}(\mathbf{z}|\mathbf{x})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right] \\&= \mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\ln p(\mathbf{x}|\mathbf{z}) \frac{p(\mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \frac{q_{\phi}(\mathbf{z}|\mathbf{x})}{p(\mathbf{z}|\mathbf{x})} \right] \\&= \mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\ln p(\mathbf{x}|\mathbf{z}) - \ln \frac{q_{\phi}(\mathbf{z}|\mathbf{x})}{p(\mathbf{z})} + \ln \frac{q_{\phi}(\mathbf{z}|\mathbf{x})}{p(\mathbf{z}|\mathbf{x})} \right] \\&= \mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} [\ln p(\mathbf{x}|\mathbf{z})] - KL[q_{\phi}(\mathbf{z}|\mathbf{x})||p(\mathbf{z})] + KL[q_{\phi}(\mathbf{z}|\mathbf{x})||p(\mathbf{z}|\mathbf{x})] .\end{aligned}$$

VAE

$$\ln p(\mathbf{x}) = \underbrace{\mathbb{E}_{\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})} [\ln p(\mathbf{x}|\mathbf{z})] - KL[q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})]}_{ELBO} + \underbrace{KL[q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}|\mathbf{x})]}_{\geq 0}$$



GAN Math

See <https://arxiv.org/pdf/1406.2661>